SOUMI PATTANAYAK

ELECTRICAL ENGINEER | POWER ELECTRONICS | ANALOG & DIGITAL CIRCUITS | RF

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PROFILE

Post graduate in Electrical Engineering seeking exciting opportunities to develop and explore in the field of Electrical and Electronics.

EDUCATION

COURSE | SPECIALIZATION	INSTITUTION	GPA | PERCENTAGE	YEAR
MASTER OF SCIENCE | ELECTRICAL ENGINEERING	The University of Texas at Dallas, TX, USA	3.455/4	2018
BACHELOR OF ENGINEERING | ELECTRICAL & ELECTRONICS ENGINEERING	Anna University (Panimalar Institute of Technology), TN, India	8.74/10	2016
STD XII	S.B.O.A. School & Junior College (C.B.S.E.), TN, India	9.5/10 (95%)	2012
STD X	S.B.O.A. School & Junior College (C.B.S.E.), TN, India	9.6/10	2010

SKILLS

HARDWARE TOOLS : Oscilloscope | Calibration Kit | Signal Generator | Vector Network Analyzer | Power Meters | Power Sources | DC & AC Motors | RLC Loads | Welding Station | Soldering Station | Buck Converter | Boost Converter.

SIMULATION TOOLS : AutoCAD Electrical | SKM System Analysis | MATLAB | Simulink | Sketch Up | PVsyst | LTspice | AWR | PSIM | Cadence Spectre | Cadence Virtuoso | ModelSim | EDI Encounter | Synopsys | DesignVision | HSPICE | Library Compiler | Waveview | PrimeTime | SiliconSmart | CATIA | AXIEM | Xilinx ISE.

SOFTWARE LANGUAGES : Verilog | C | C++ | JAVA.

ACADEMIC EXPERIENCE

Power Electronics Lab Experience	-Performed tests on Buck converter (PMLK BUCK-LM3475, TPS54160) and studied about their performance dependencies on various parameters and behavior variation during disturbances. -Observed impact of operating conditions on efficiency, impact of passive devices and switching frequency on current and voltage ripples, impact of cross-over frequency and passive devices on load transient response.

RF Lab Experience	-Designed and simulated various microwave components using Microwave Office AWR and AXIEM EM Simulator. Fabricated components on Duroid 5800 and FR-4 substrates. -Tested and validated components using power meter, Vector Network Analyzer, Spectrum Analyzer. -The designed microwave components include dipole antenna, amplifier, resonator, power divider, directional coupler, low pass filter, bandpass filter.
Design of Low Noise Amplifier	-Designed a low noise amplifier with two transistors in cascode mode -Performed noise and power analysis to obtain low power consumption of 2.4mW and noise figure less than 2.5dB
Compact Dual Band Wilkinson Divider with Wide Frequency Ratio	-Designed and analyzed a conventional Wilkinson power divider -Designed and analyzed a dual band Wilkinson power divider with lumped elements operating at 0.5GHZ and 2GHz to achieve better isolation and wide frequency of operation
Bang-Bang Control of Buck Converter	-Designed a buck regulator with input voltage of 2.7V-4.2V using MOSFET switches in Cadence -Performed output regulation by designing a control loop for bang-bang control of the converter -Designed high gain comparator, SR Latch, buffer and dead time circuitry for the feedback loop
IBC design for High Voltage Applications	-Designed a two stage Interleaved Boost Converter connected in cascade -Converted low input voltage to high output voltage that can be supplied to high power applications -Simulation results provided an output voltage of approximately 535V
Design of 32 bit Arithmetic & Logical Unit	-Performed behavioral coding of 32 bit Arithmetic & Logical Unit (ALU) with basic arithmetic operations, logical functions, and shift operations using Verilog -Verified and tested the code using ModelSim and generated cell report using DesignVision. Total number of cells was 6969 -Designed general logic gates and falling-edge D flip flop (schematic and layout) in Cadence -Verified logic gates using HSPICE and generated cell library using SiliconSmart -Performed automatic placing and routing using EDI Encounter and static timing analysis using PrimeTime
Differential Amplifier for Audio Amplification	-Designed an amplifier with differential input and single-ended output for audio amplification -Performed gain and power analysis to achieve high gain and good power supply rejection
Control Mechanisms and Stability Analysis of DC-DC Converter	-Designed buck and boost converters with resistive load and constant power load -Performed minimal phase and non-minimal phase analysis for generalized average model design of the converters -Designed feedback control circuits for the dc-dc converters -Analyzed control techniques such as P control, PI control, sliding mode control, and feedback linearization

On- & Off- Grid system design using Real Time data	Site: Building of Electrical and Electronics Engineering, Panimalar Institute of Technology, Chennai, India. -Built a measuring station using monocrystalline PV panels on-site to record irradiation data for a span of six months -Collected radiation data from National Institute of Wind Energy (NIWE) and National Renewable Energy Laboratory (NREL), India to analyze and compare with measured Global Horizontal Irradiance (GHI) values for accuracy and deviation -Performed shadow analysis for the site to estimate utilizable unshaded region using Google SketchUp -Designed PV system for the utilizable area of the site using CATIA -Simulated and estimated power outputs for on-grid and off-grid system using PVsyst and performed cost analysis for the site
Publication: "Grid Connected Photovoltaic System based on the Switched Coupled – Inductor Quasi-Z-Source Inverter"; IJAREEIE; Volume 4, Issue 7, July 2015; ISSN 2278 – 8875 (online)	
Publication: "A Novel Analysis and Control of Matrix Converter Using Mathematical Model Under Distorted Input Voltage Conditions"; Australian Journal of Basic and Applied Sciences, Vol. 10(1), Pages: 165-173, January 2016	

INTERNSHIPS AND IN-PLANT TRAININGS

- SIEMENS LTD.: Internship on drives and motors, PLC programming software (25.05.2015 to 08.06.2015)
- BGR ENRGY SYSTEMS LTD.: Internship on power distribution and AC systems (04.06.2014 to 17.06.2014)
- Kuttalam Gas Turbine Power Station, Maruthur (22.12.2014 to 26.12.2014)
- Breaks India Limited (28.04.2014 to 29.04.2014)

PERSONAL DETAILS

Father's Name : Satyabrata Pattanayak
Date of Birth : 24.10.1994
Languages known : English, Hindi, Bengali, Tamil
Leisure activities : Drawing, Music and involving in volunteering activities
Contact address : C32, Metropolitan apts., Rajankuppam, Vanagaram-Ambattur Rd, Ayanambakkam, Chennai-600095

ACHIEVEMENTS AND ACTIVITIES

- Awarded "Certificate of Merit" for attaining 40th rank in Anna University, among 15141 graduate students under the Faculty of Electrical Engineering.
- Awarded scholarship for Science and Arts stream by the Central Board of Secondary Education, Govt. of India for being among the top 2% performers in the country.
- Awarded Cambridge English: Business Preliminary (BEC Preliminary) certificate
- Volunteered with Chennai Volunteers (2014-2016)

References available upon request. All information in this document is accurate and true to the best of my knowledge.
Soumi Pattanayak